TELIPORT
Script Your Future
Season 4

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Case Study No. 3

Team Details

Team MM

Manas Girdhar (Team Leader)

Meenakshi Satpathy

Birla Institute of Technology, Mesra

THINK. BUILD. CREATE. WIN

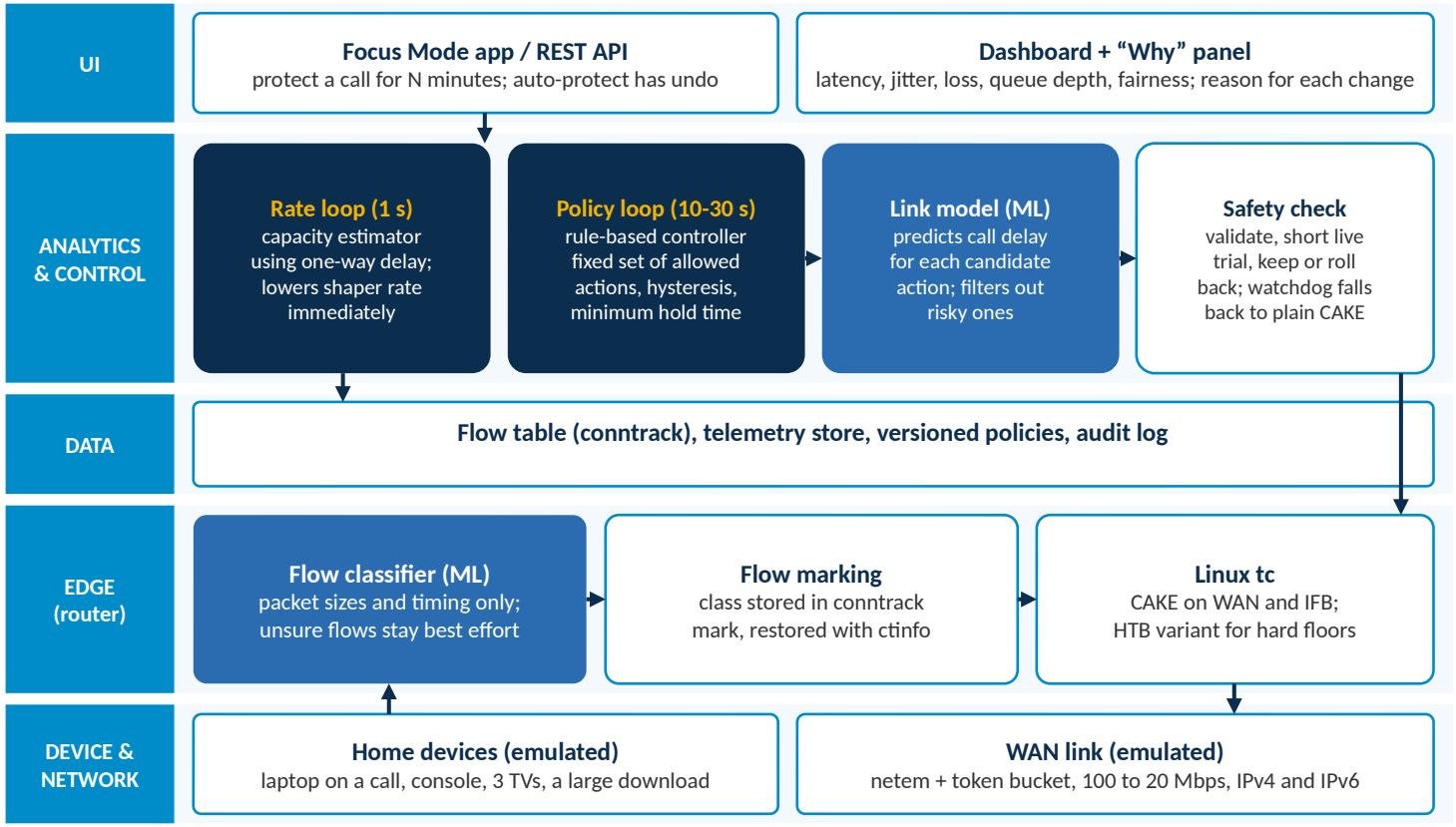
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PulseQoS: the call never notices the download

An adaptive QoS controller for home broadband, built on Linux tc and CAKE. No payload inspection.

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System architecture



Key design decisions

- **Tins:** call audio and games go to the Voice tin, call video to the Video tin (Voice is capped at 25% in diffserv4)
- **Trust:** DSCP marks coming from the Internet are cleared, so outside traffic cannot claim priority
- **Two loops:** a speed drop is handled within a second; policy changes go through the safety check
- **Privacy:** no payload or domain logging; the intent API is LAN-only and needs a token

Where we use ML, and how we check it helps

1. Flow classifier: LightGBM on flow features with conformal prediction. A flow is prioritised only when the model is confident, which keeps wrong boosts under a target rate (say 1%). **Compared with port-based rules.**

2. Link model: trained offline on lab runs, it predicts how a policy change will affect call delay, so risky changes are dropped before the live trial. **Measured by seconds of bad call delay avoided.**

3. Auto-protect: when a call starts on a busy link it is protected automatically, within a daily per-device limit, and the user can undo it. **Measured by time from call start to protection.**

Each ML part is tested against the plain rule-based version. If it does not improve the numbers, we drop it.

Early test we ran (27 Oct)

95th percentile ping delay during a download (ms)



Download speed was the same in both cases (18.7 of 20 Mbps), so the call was protected without starving the download.
Setup: 2 Linux namespaces, 20 Mbit link, 4 TCP streams, 50 pings/s, 5 runs each. No WAN delay was emulated. In Round 2 we will compare against CAKE, CAKE autorate and cake-autorate across 14 scenarios.

Scope: the prototype runs in emulation. A real router would also need hardware offload handling, enough CPU for gigabit shaping, and Wi-Fi airtime fairness.